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Dunn et al.

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(54) **OPTIMIZED SLOT MOTOR FOR REMOTE
SECONDARY CONTACTS IN A CIRCUIT
BREAKER**

USPC 335/16
See application file for complete search history.

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H01H 71/52 (2006.01)

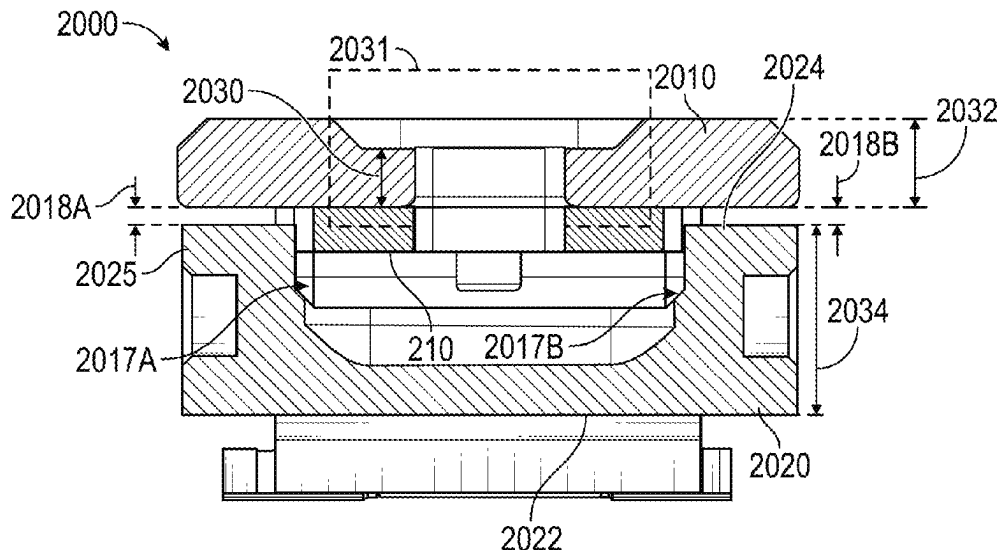
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(2013.01)

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CPC H01H 77/108; H01H 71/523; H01H 1/54;
H01H 89/06

(57) **ABSTRACT**

A slot motor for use with secondary contacts in a circuit breaker includes a top slot motor component structured to be attached to a moving arm of the secondary contacts; and a U-shaped bottom slot motor component including a base and a pair of legs extending upward from the base, the U-shaped bottom slot motor component structured to be separated from the top slot motor component by vertical gaps between the top slot motor component and ends of the pair of legs, wherein the slot motor is structured to generate a magnetic field producing a force to maintain the secondary contacts in closed position upon passing high current through the moving arm.

13 Claims, 19 Drawing Sheets



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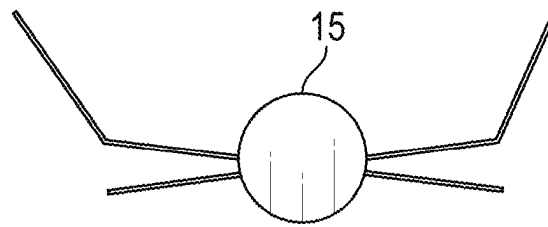


FIG. 1A

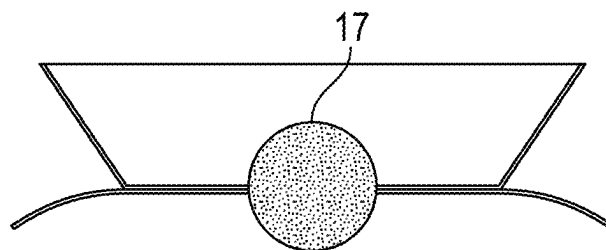


FIG. 1B

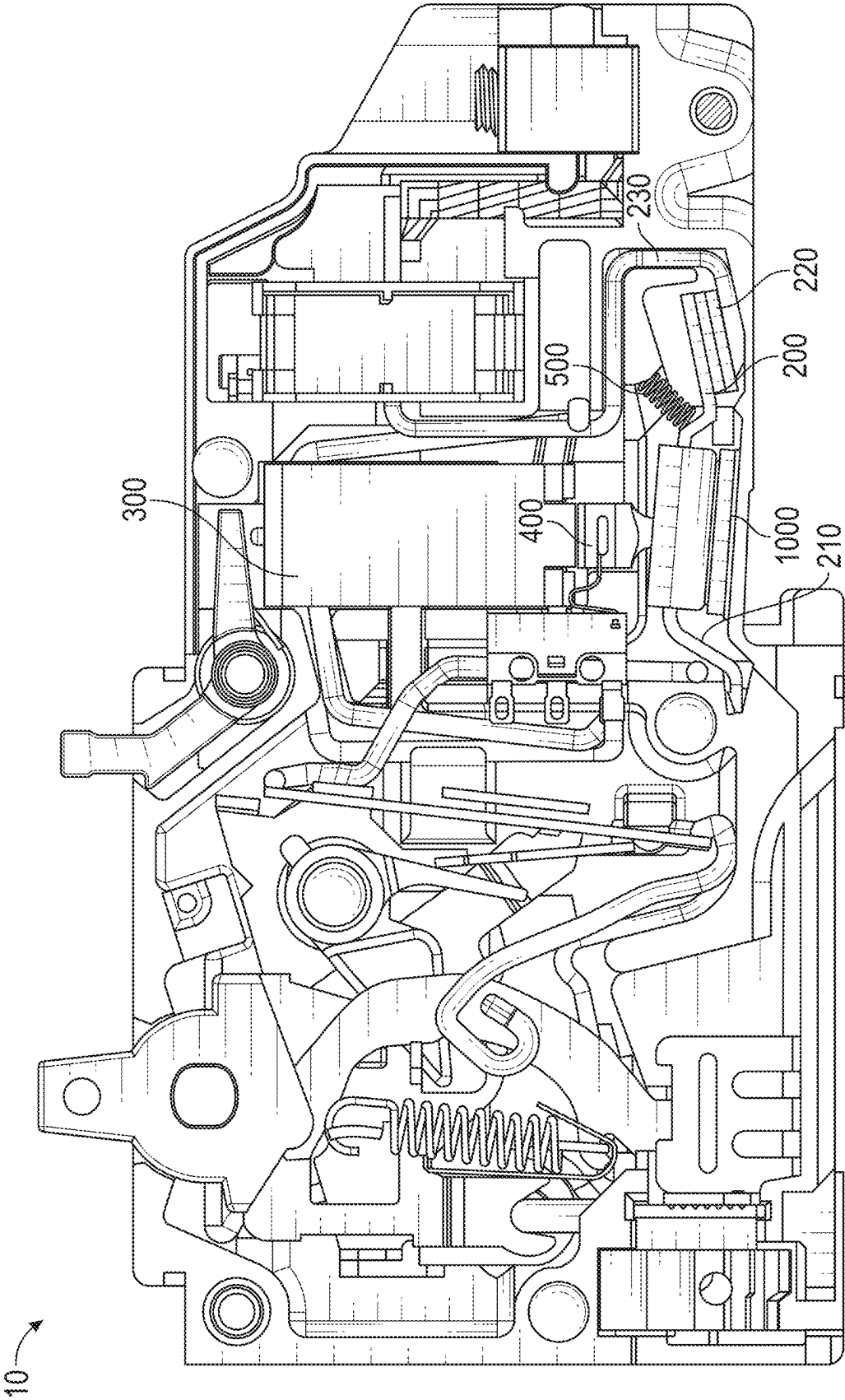
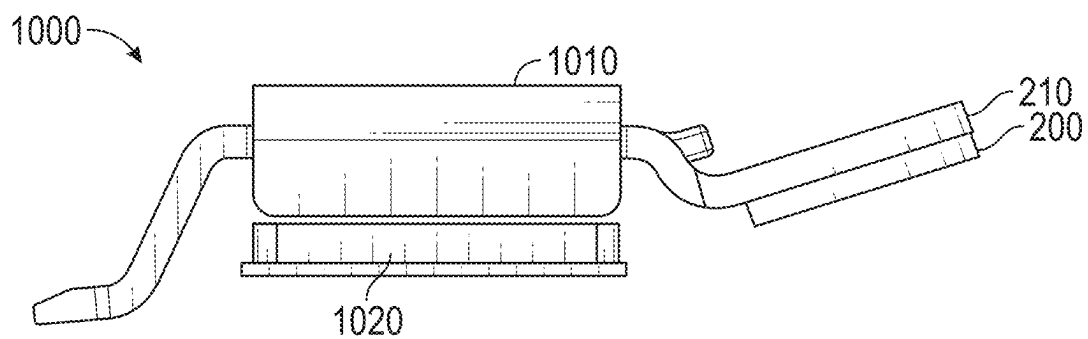
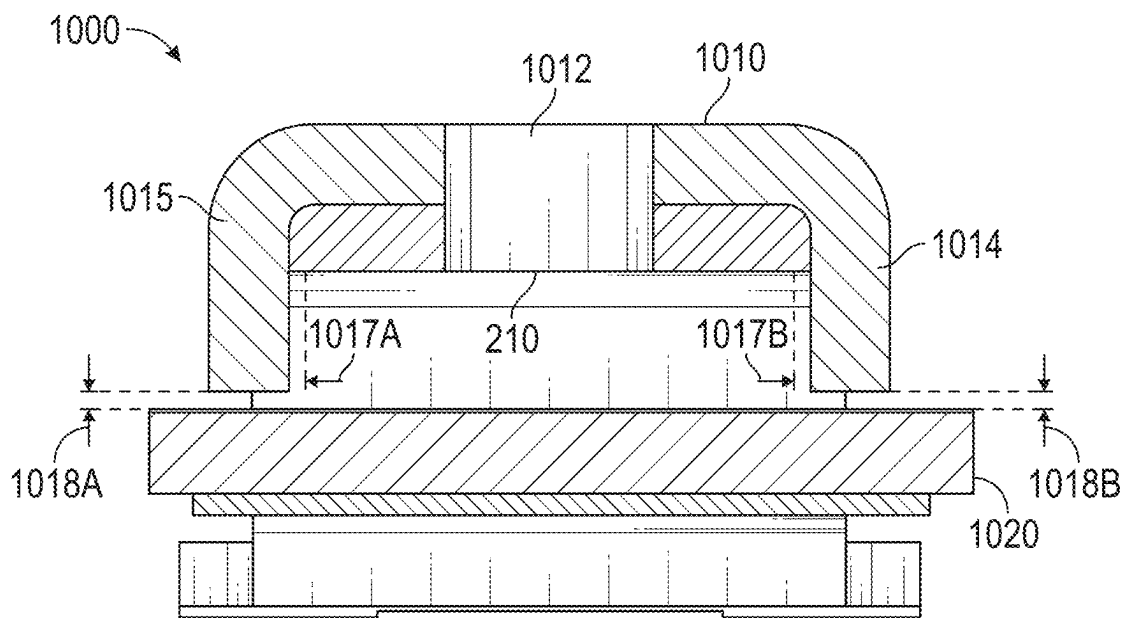


FIG. 2 (Prior Art)



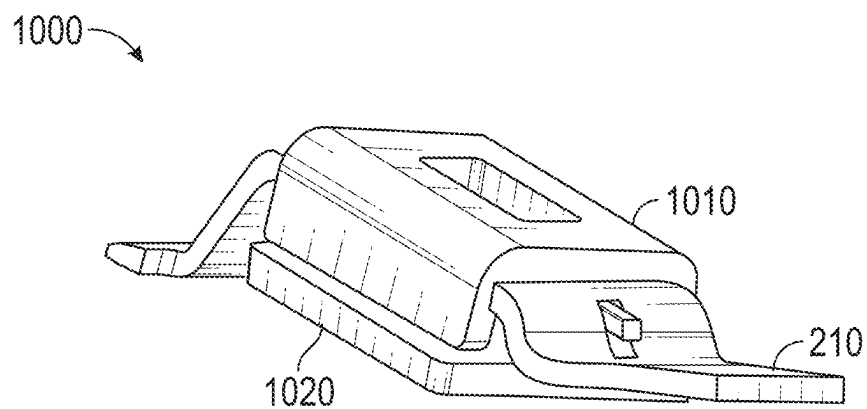


FIG. 3C (Prior Art)

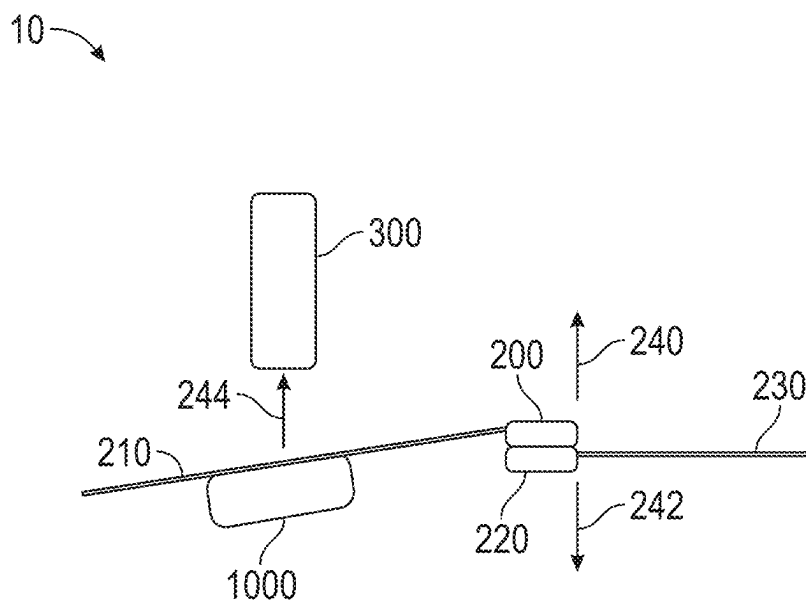
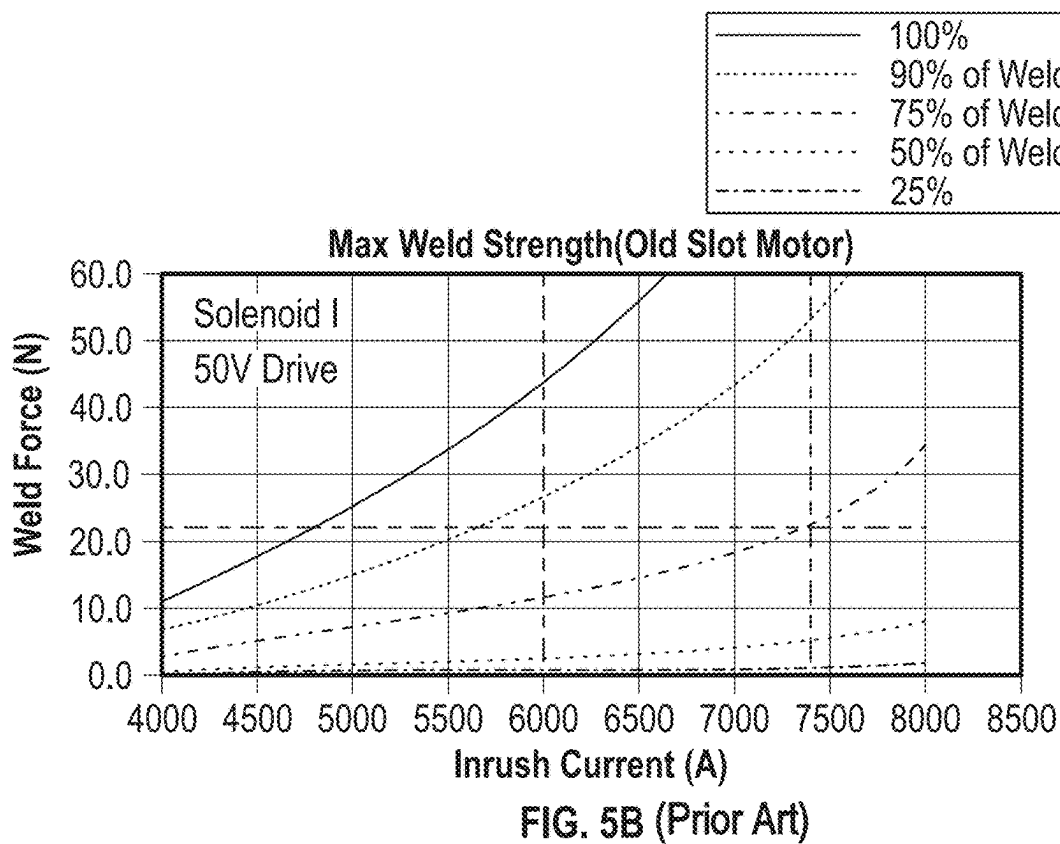
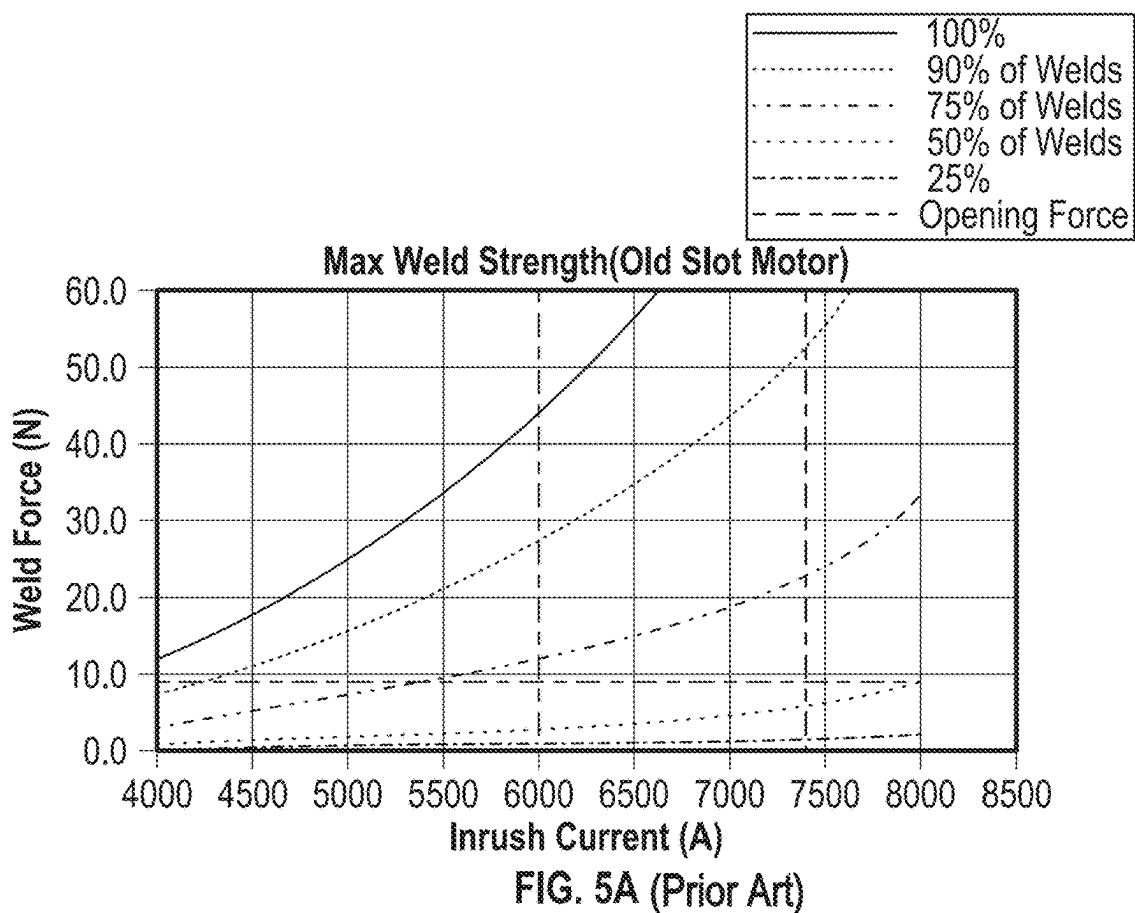


FIG. 4 (Prior Art)



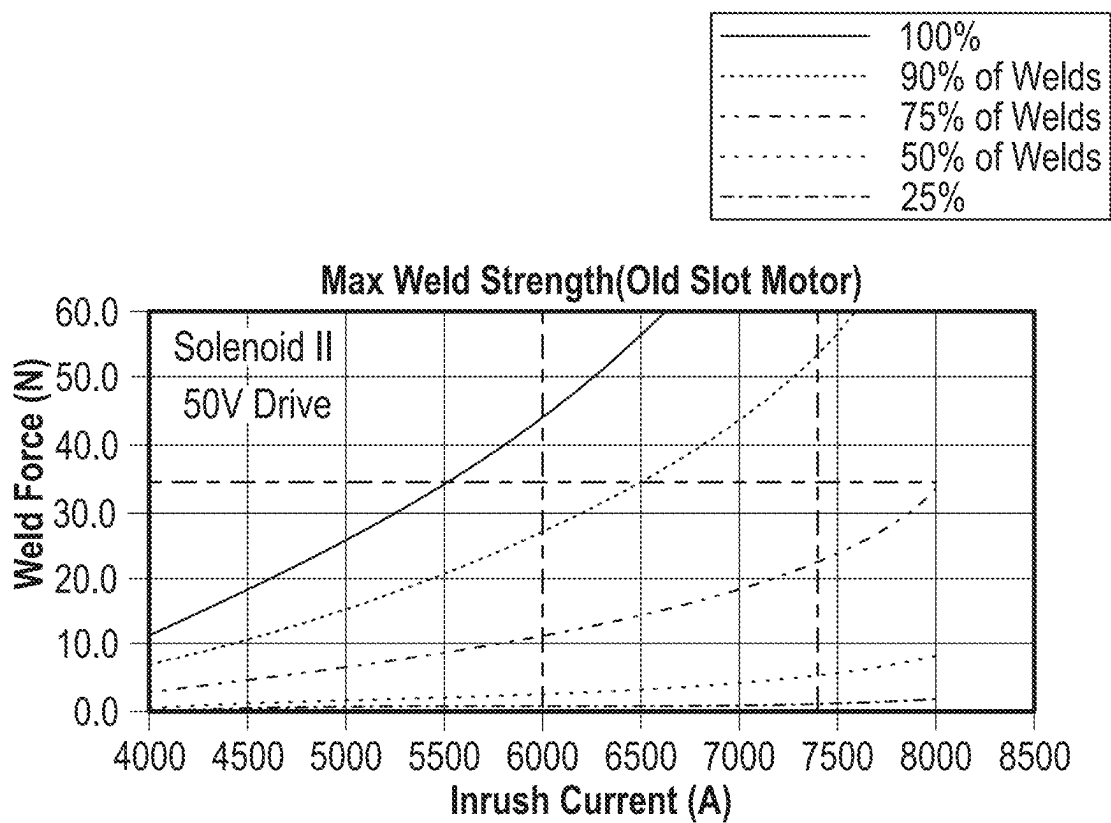
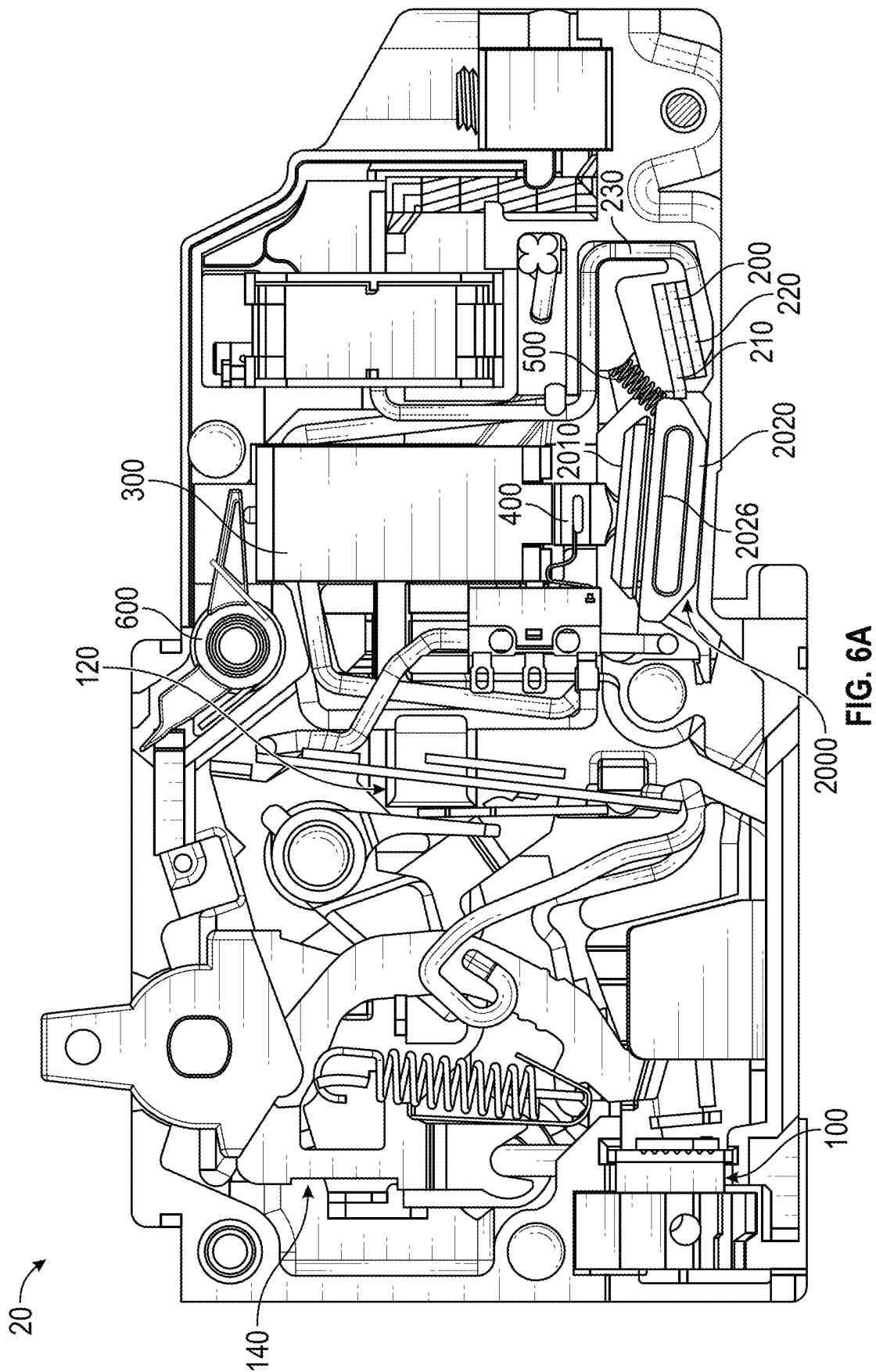


FIG. 5C (Prior Art)



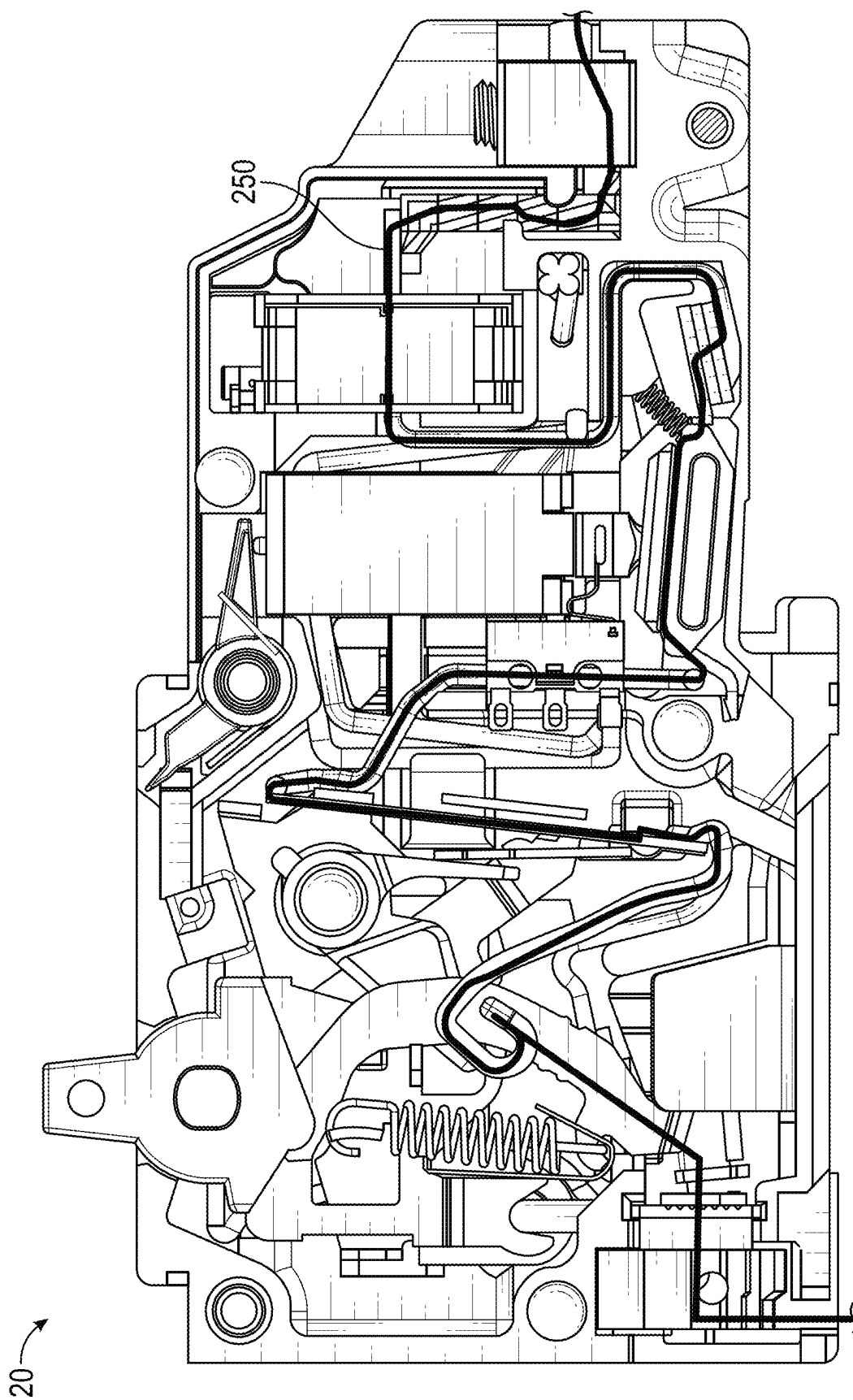


FIG. 6B

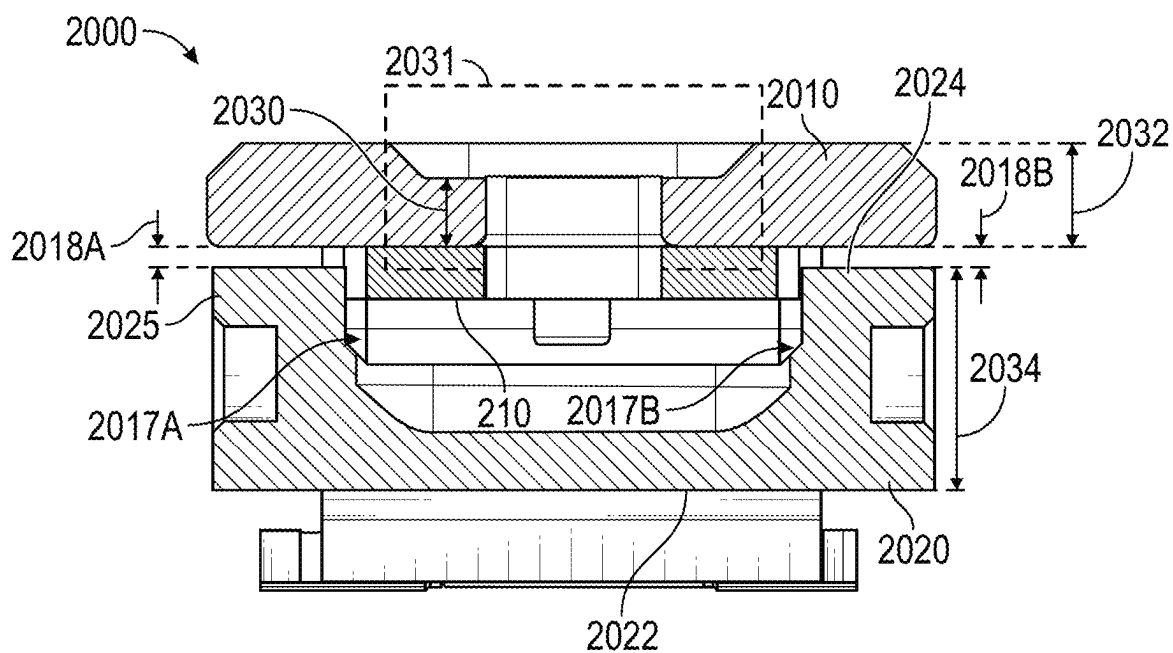


FIG. 7A

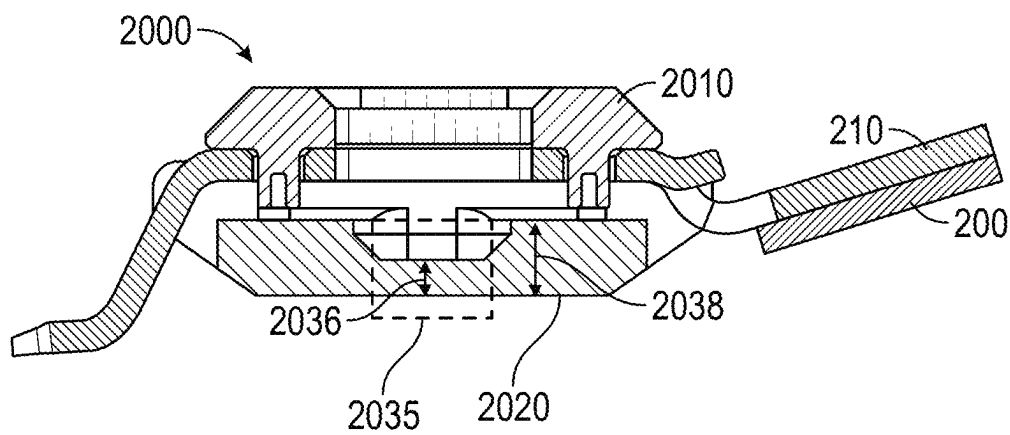


FIG. 7B

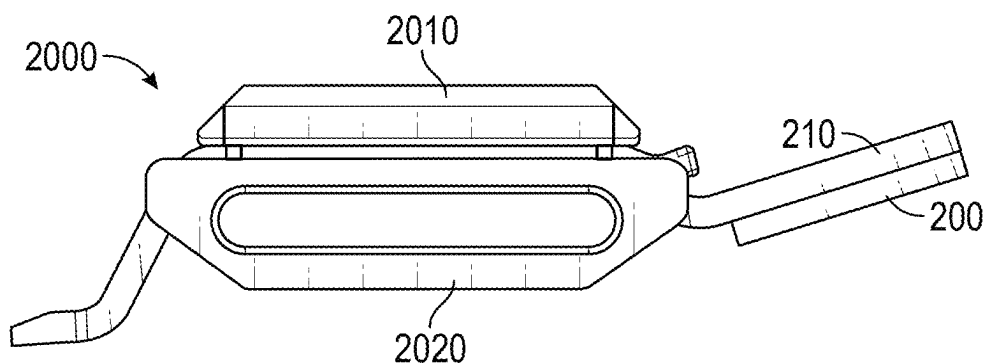


FIG. 7C

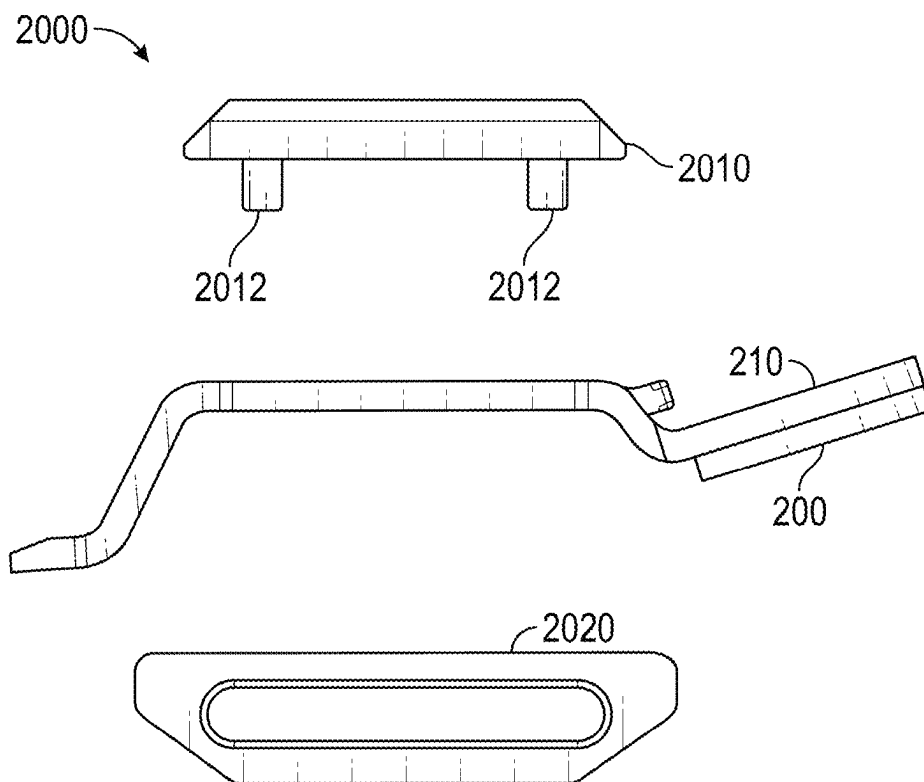


FIG. 7D

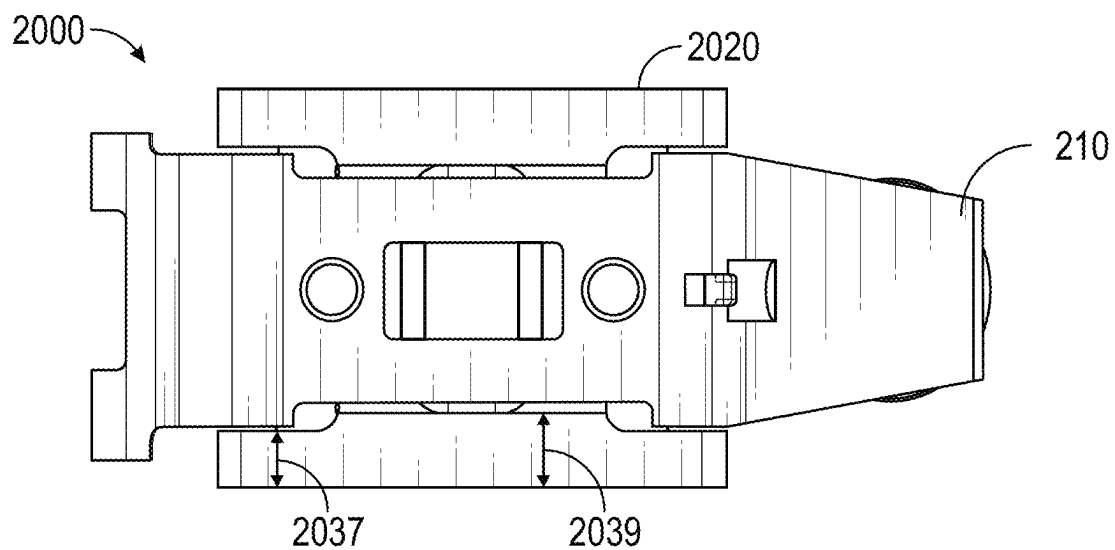


FIG. 7E

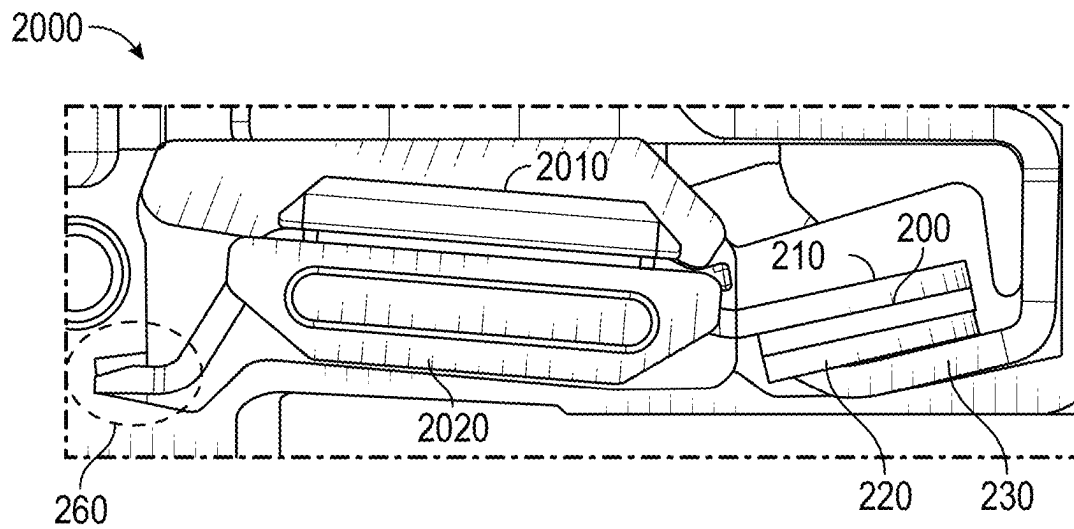


FIG. 8A

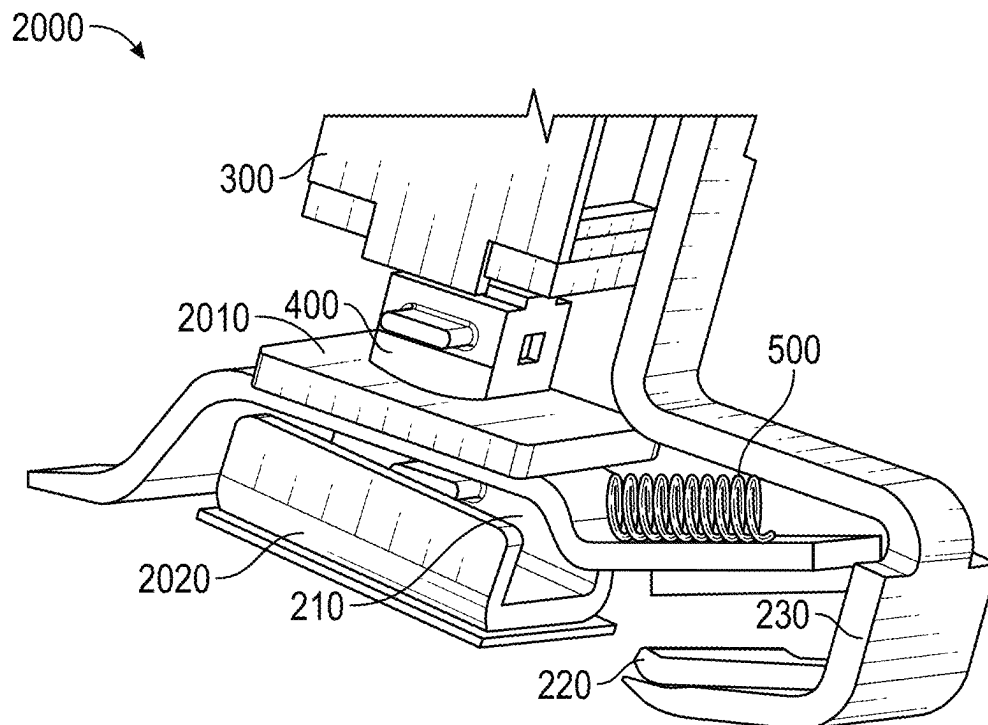


FIG. 8B

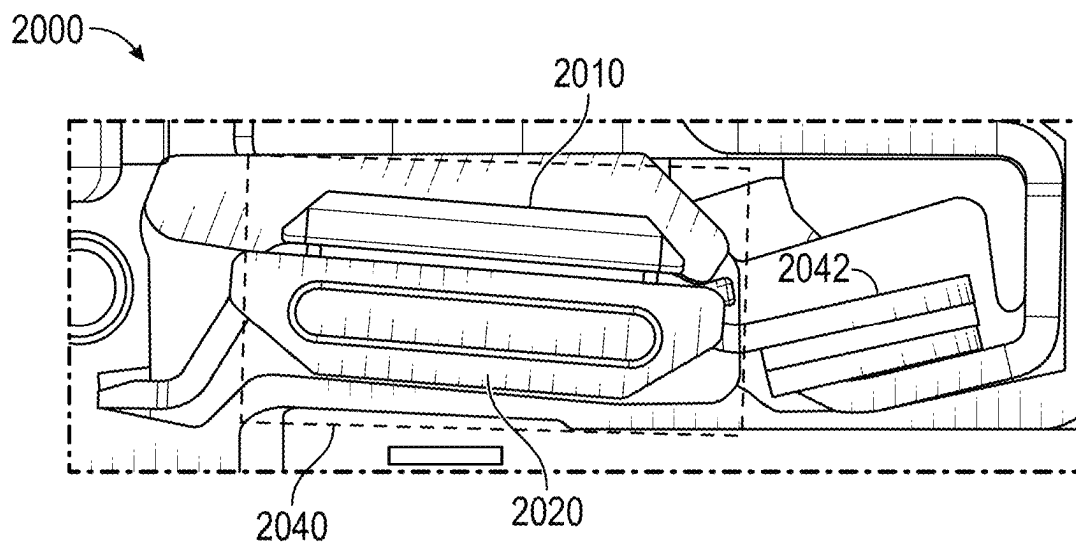


FIG. 9A

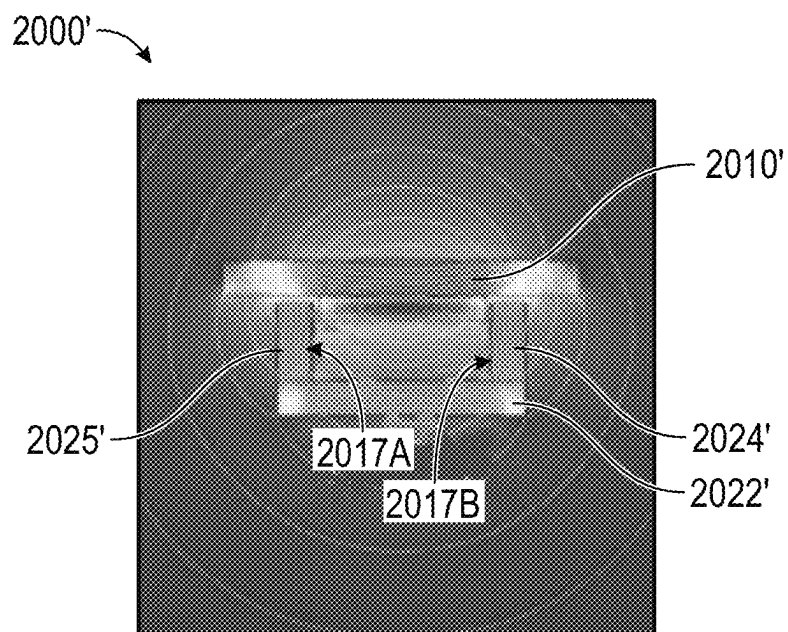


FIG. 9B

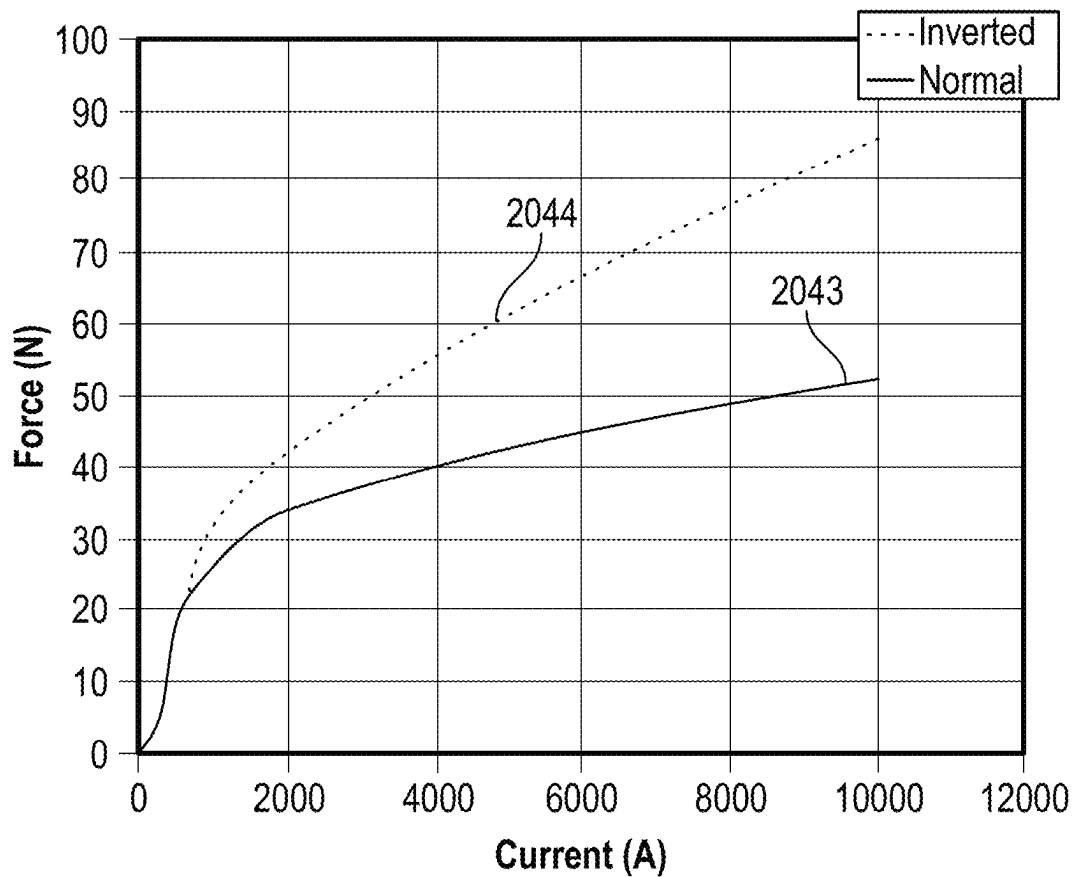


FIG. 9C

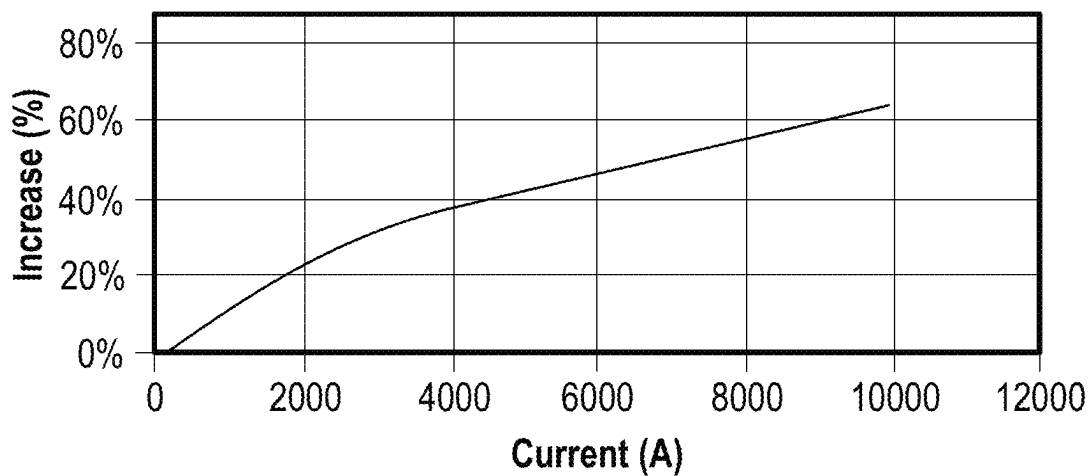


FIG. 9D

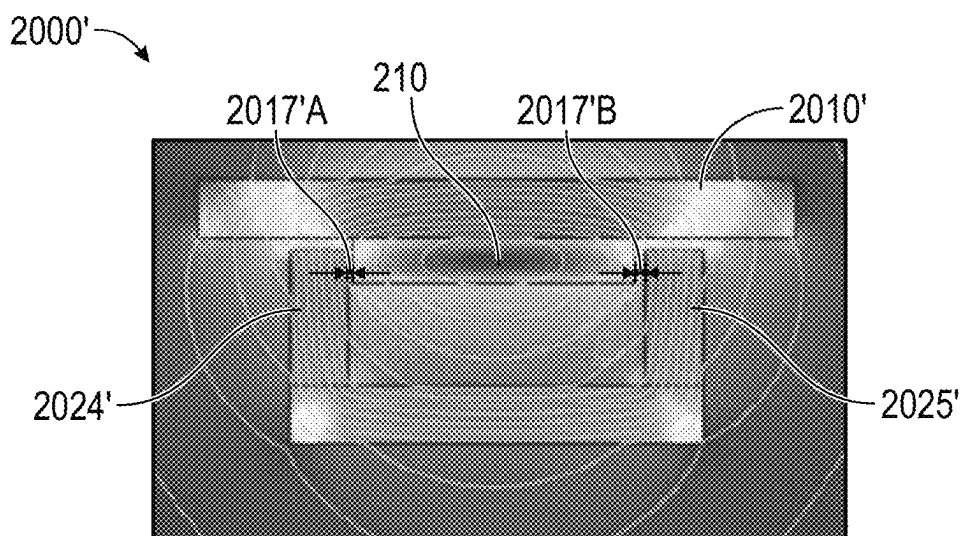


FIG. 10A

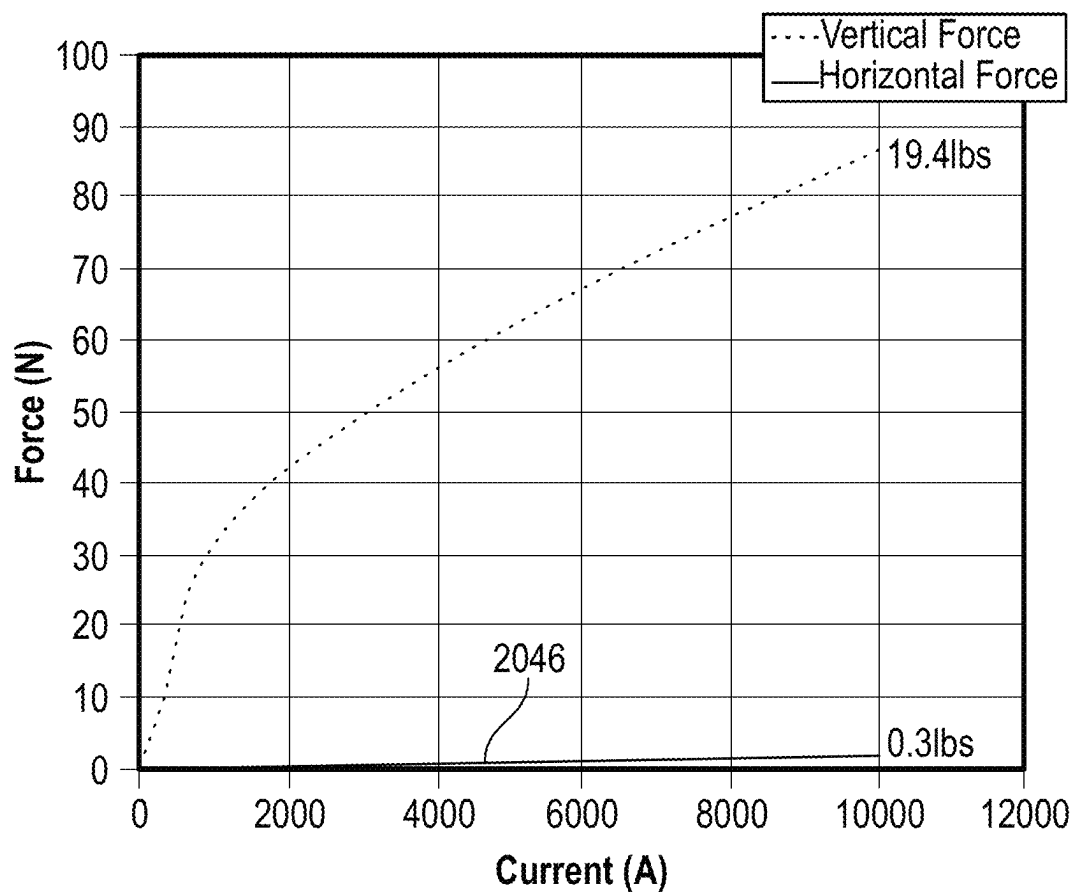


FIG. 10B

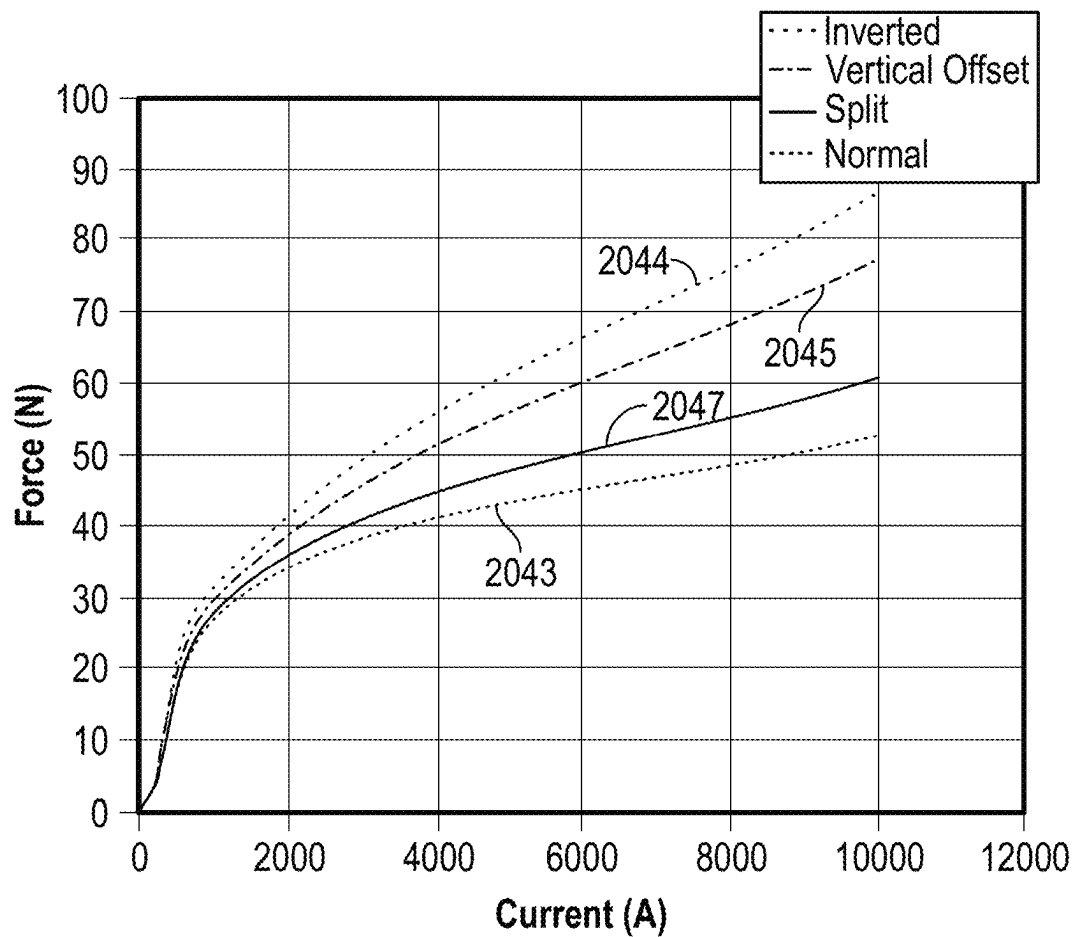


FIG. 11A

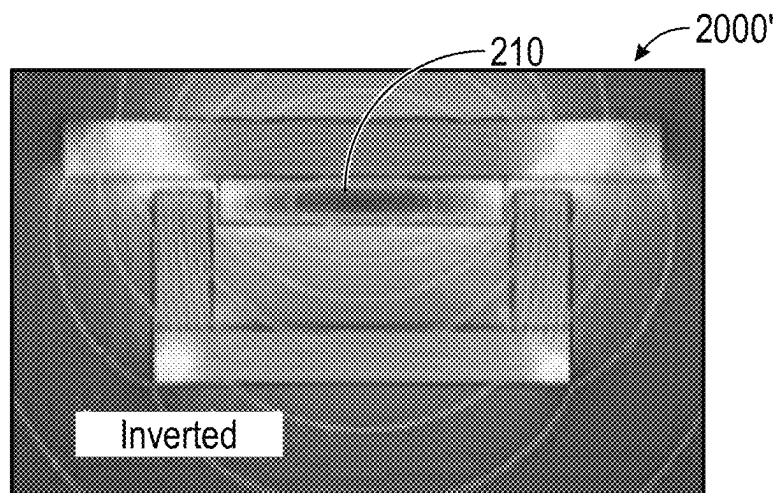


FIG. 11B

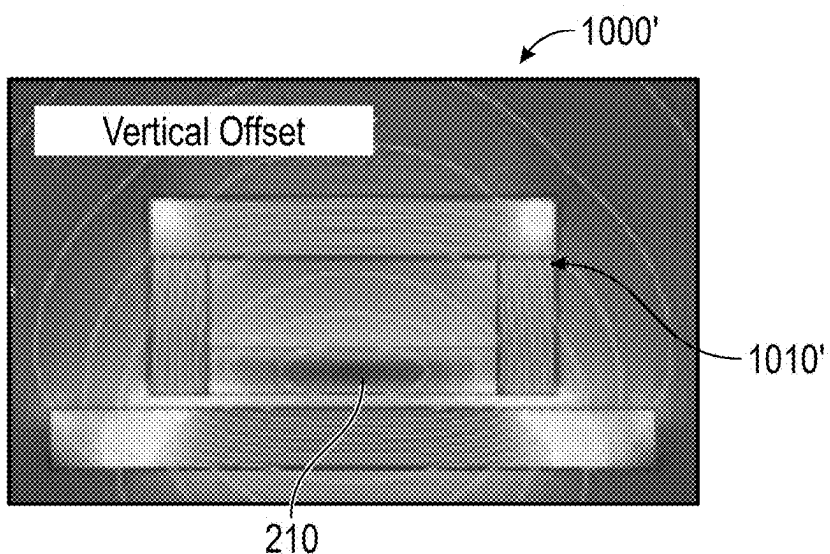


FIG. 11C

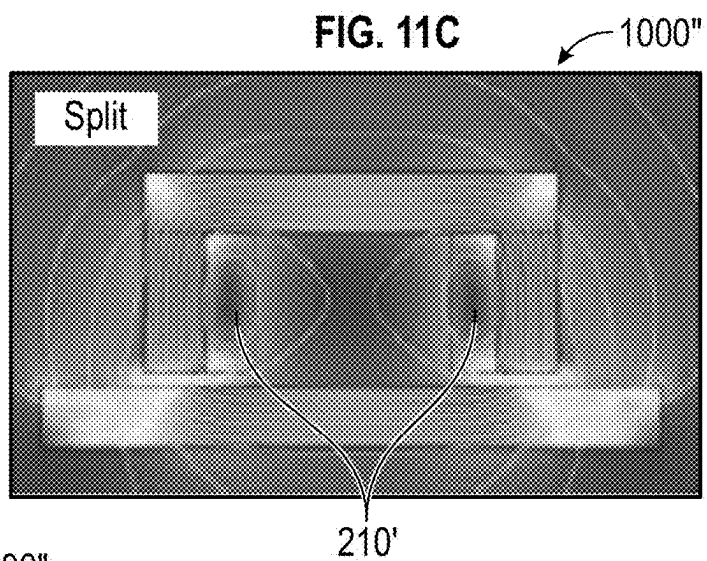


FIG. 11D

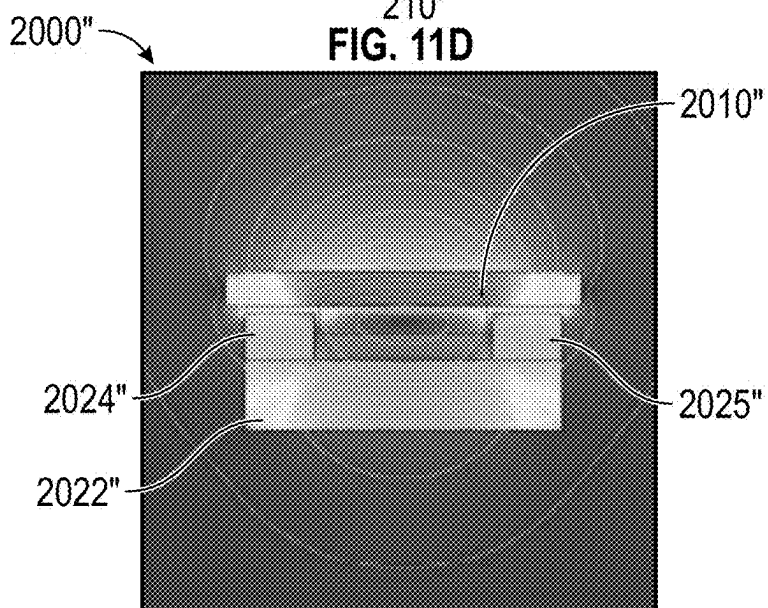


FIG. 12A

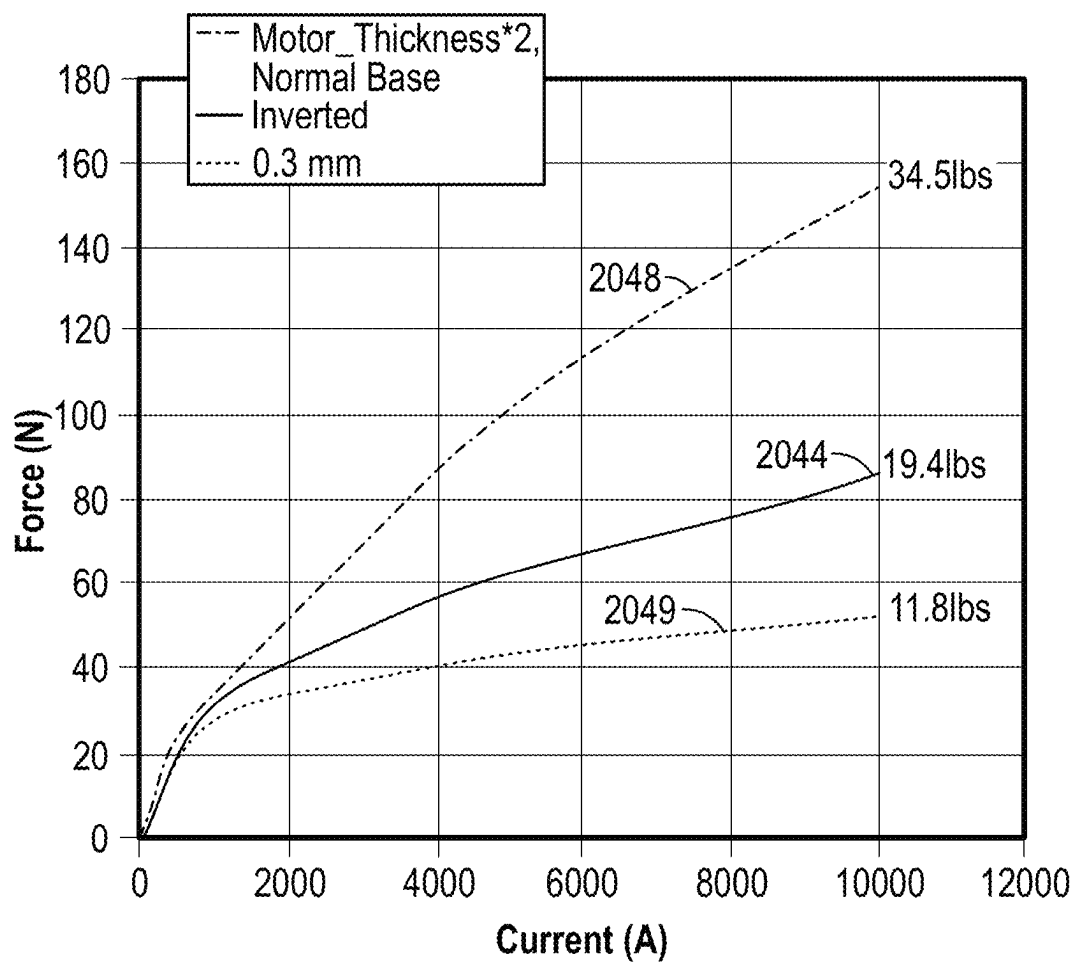


FIG. 12B

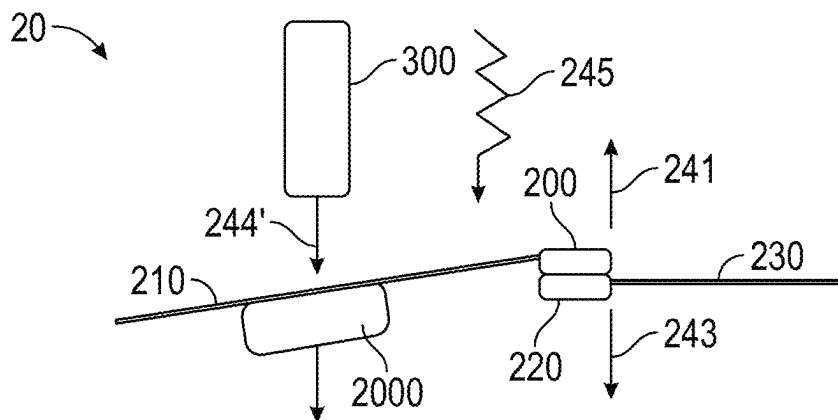


FIG. 13A

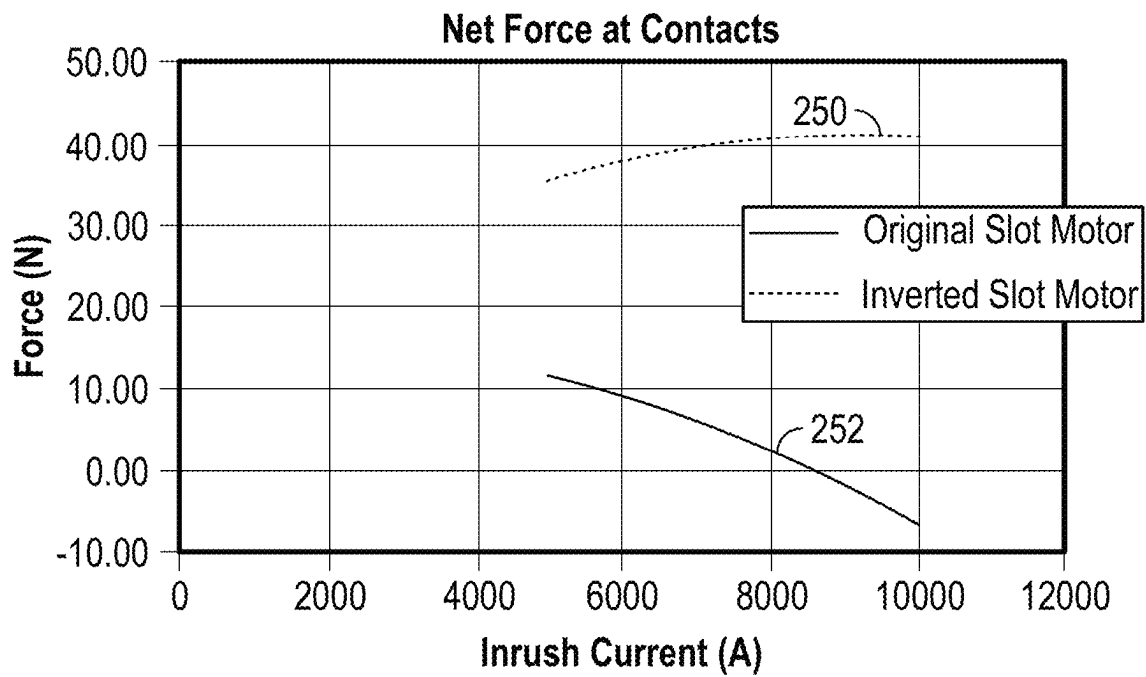


FIG. 13B

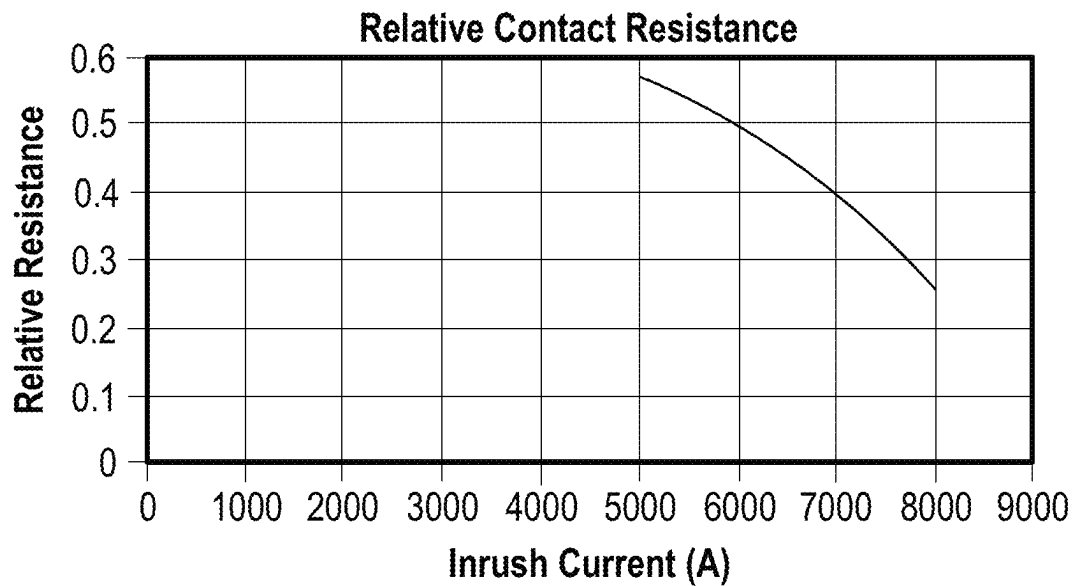


FIG. 13C

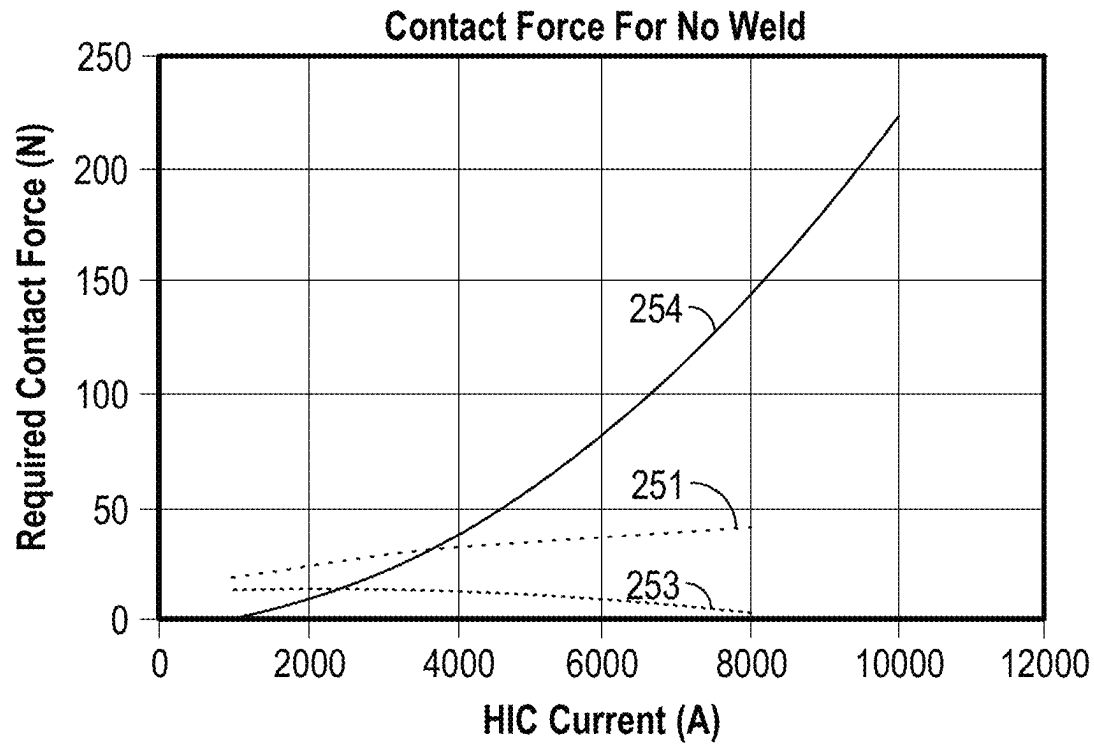


FIG. 14

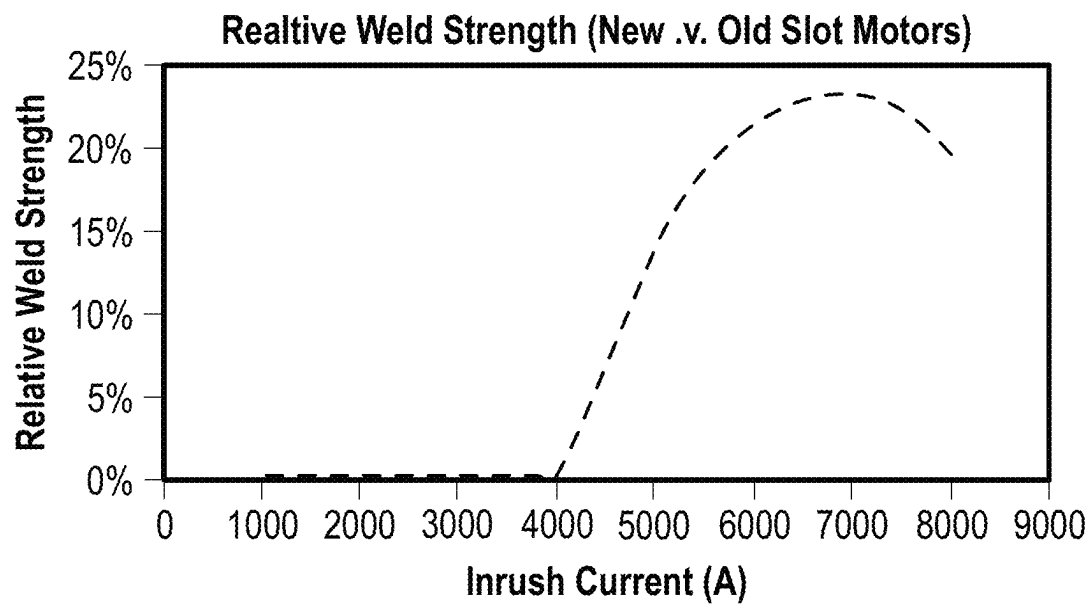


FIG. 15

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OPTIMIZED SLOT MOTOR FOR REMOTE SECONDARY CONTACTS IN A CIRCUIT BREAKER

CROSS REFERENCE TO RELATED APPLICATIONS

The present application claims priority to U.S. Provisional Patent Application No. 63/289,772 filed Dec. 15, 2021, which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The disclosed concept relates generally to a slot motor, and in particular an optimized slot motor for remote secondary contacts in a circuit breaker.

BACKGROUND OF THE INVENTION

During a short circuit event, secondary contacts in remote-controlled circuit breakers (e.g., smart circuit breakers controllable by an end-user via wireless or wired connections) may rely on a slot motor to prevent chattering and excess heat generation, which can lead to tack or contact welding at the contacts.

Typically, one component of the slot motor is attached to a moving arm of the secondary contacts, while the other component is fixed in place within the assembly. The magnetic force between the two keeps the contacts closed during a high current event. However, the contacts still open frequently at low voltage drive and at a very high rate at high voltage drive, evidencing inefficacy of the conventional slot motor. Further, sometimes a circuit breaker with the conventional slot motor has sufficient contact force to prevent welds at inrush current smaller than a threshold (e.g., approximately 2500 A), allowing the contacts to open and tack weld to occur in the inrush current events beyond such threshold (e.g., approximately 2500 A). Upon the occurrence of the weld, the secondary contacts become useless as they are now welded together.

There is a considerable room for improvement in the slot motors for the secondary contacts in circuit breakers.

SUMMARY OF THE INVENTION

These needs, and others, are met by embodiments of the disclosed concept in which a slot motor for use with secondary contacts in a circuit breaker is provided. The slot motor includes a top slot motor component structured to be attached to a moving arm of the secondary contacts; and a U-shaped bottom slot motor component including a base and a pair of legs extending upward from the base, the U-shaped bottom slot motor component structured to be separated from the top slot motor component by vertical gaps between the top slot motor component and ends of the pair of legs, where the slot motor is structured to generate a magnetic field producing a force to maintain the secondary contacts in a closed position during a high current event.

Another embodiment provides a circuit breaker connected to a power source via a line conductor and a load via a load conductor. The circuit breaker includes: primary contacts coupled to the line conductor; an operating mechanism structured to cause the primary contacts to trip open the circuit breaker during a high current event; secondary contacts coupled to the load conductor and structured to open or close the circuit breaker based on a user instruction received upon tripping of the circuit breaker; and a slot motor coupled

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to a secondary moving arm of the secondary contacts. The slot motor includes a top slot motor component structured to be attached to a moving arm of the secondary contacts; and a U-shaped bottom slot motor component including a base and a pair of legs extending upward from the base, the U-shaped bottom slot motor component structured to be separated from the top slot motor component by vertical gaps between the top slot motor component and ends of the pair of legs, where the slot motor is structured to generate a magnetic field producing a force to maintain the secondary contacts in a closed position during a high current event.

BRIEF DESCRIPTION OF THE DRAWINGS

A full understanding of the invention can be gained from the following description of the preferred embodiments when read in conjunction with the accompanying drawings in which:

FIGS. 1A-B illustrate a weld and a melt spot created by an inrush current going through secondary contacts in a circuit breaker.

FIG. 2 illustrates a circuit breaker with a conventional slot motor for secondary contacts;

FIGS. 3A-C illustrate an example conventional slot motor for secondary contacts;

FIG. 4 illustrates opening strength of the secondary contacts using an example conventional slot motor;

FIGS. 5A-C illustrate maximum weld strengths using example conventional slot motors;

FIGS. 6A-B illustrate internal views of a circuit breaker with a slot motor according to an example embodiment of the disclosed concept;

FIGS. 7A-E illustrate a slot motor in accordance with an example embodiment of the disclosed concept;

FIGS. 8A-B illustrate a slot motor according to an example embodiment of the disclosed concept;

FIGS. 9A-D illustrate magnetic fields and contact force generated by a slot motor according to an example embodiment of the disclosed concept;

FIGS. 10A-B illustrate contact force generated based on horizontal misalignment of the moving arm according to an example embodiment of the disclosed concept;

FIGS. 11A-D illustrate magnetic fields and contact force generated by a plurality of conventional slot motors and an inventive slot motor according to an example embodiment of the disclosed concept;

FIGS. 12A-B illustrate magnetic fields and contact force generated by a slot motor according to an example embodiment of the disclosed concept;

FIGS. 13A-C illustrate balance of forces at the secondary contacts in a circuit breaker according to an example embodiment of the disclosed concept;

FIG. 14 illustrates contact forces for no weld according to example embodiments of the disclosed concept; and

FIG. 15 illustrates relative weld strength of a slot motor according to an example embodiment of the disclosed concept.

DETAILED DESCRIPTION OF THE INVENTION

Directional phrases used herein, such as, for example, left, right, front, back, top, bottom and derivatives thereof, relate to the orientation of the elements shown in the drawings and are not limiting upon the claims unless expressly recited therein.

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As employed herein, the statement that two or more parts are “coupled” together shall mean that the parts are joined together either directly or joined through one or more intermediate parts.

In general, during short circuit events secondary contacts in remote-controlled circuit breakers (e.g., smart circuit breakers controllable by an end-user via wireless or wired connections) may rely on a slot motor to prevent chattering and excess heat generation, which can lead to tack or contact welding at the contacts. Contact welding occurs when there is enough energy generated during an inrush current event to melt the material of the contacts. There are two possible sources of melt energy: contact resistance causing resistance heating and arc created when the contacts are not touching due to, e.g., insufficient force to hold the contacts together during the short circuit event. An arc flash may result if the slot motor grounds or if a slot motor gap is large. Contact resistance R is provided as:

$$R = \frac{\rho}{2a} \quad \text{EQ. 1}$$

where ρ is material resistivity of the contacts and a is area in contact. Assuming asperities on one contact surface touch the other contact surface, the area a in contact depends on hardness and contact force, as follows:

$$a = \sqrt{\frac{F}{\pi H}} \quad \text{EQ. 2}$$

where F is contact force and H is hardness. Contact resistance R can be also provided as:

$$R = \frac{\rho}{2} \sqrt{\frac{\pi H}{F}} \quad \text{EQ. 3}$$

As such, the bigger the contact force F , the smaller the contact resistance R . The resistance heating E_{heating} in the contact caused by the contact resistance can be provided as:

$$E_{\text{heating}} = RI^2 \Delta t \quad \text{EQ. 4}$$

where I is the inrush current. For current below the minimum weld current, there is not enough heating to reach the melting point. The energy E_{melt} used for melting the contact material can be:

$$E_{\text{melt}} = R(I^2 - I_{\text{min weld}}^2) \Delta t \quad \text{EQ. 5}$$

where $I_{\text{min weld}}$ is the minimum current for a weld to form. In a high inrush current event, the current I going through the contact resistance generates heat. The melt energy E_{melt} deposited in an arc flash is many times greater than the melt energy E_{melt} when the contacts are closed. If the contacts open during the high inrush current event, then very strong welds (e.g., 711.7 N (i.e., 160 lbf)) welds can be expected. As such, if there is sufficient heat, it will melt the contact materials and create a contact weld **15** as shown in FIG. **1A**. The volume v_{melt} of melted material is:

$$v_{\text{melt}} \approx \frac{E_{\text{melt}}}{d(C_v T_{\text{melt}} + C_{\text{fusion}})} \quad \text{EQ. 6}$$

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where d is the density, C_v is the specific heat of the contact material, C_{fusion} is the latent heat of fusion of the contact material, and T_{melt} is the melting temperature of contact materials. The area a_{melt} of the melt spot **17** as shown in FIG. **1B** from simple geometry is:

$$a_{\text{melt}} = \pi \left(\frac{3v_{\text{melt}}}{4\pi} \right)^{\frac{2}{3}} \quad \text{EQ. 7}$$

The strength F_{weld} of the weld is given by the area of the melt spot and the material tensile strength as follows:

$$F_{\text{weld}} \approx \frac{a_{\text{melt}}}{\sigma} \quad \text{EQ. 8}$$

where σ is the tensile strength. The weld strength F_{weld} is a function of the deposited energy and can be provided as:

$$F_{\text{weld}} \approx k(E_{\text{melt}})^{\frac{2}{3}} \quad \text{EQ. 9}$$

where K is constant.

The practical problem is to get a good value for the constant K and this depends on the contact material. Experimental values are better than theoretically calculated ones. A large variation in weld strengths (e.g., 10:1) has been observed. Experimental measurements of variation of weld strength show that the majority of welds (80%) are only a small fraction (30% or less) of the maximum weld strength. In practice, this means a few high inrush current events result in “super welds.” Most inrush current events result in low strength tack welds.

The area of contact theory and contact welding is an active area of research and substantial work on these subjects has been undertaken. It has been shown that it is possible to estimate the minimum current required to weld two pieces of metal together. The minimum current required to weld i_w depends on resistivity, thermal conductivity, hardness, contact force as shown below:

$$i_w = \frac{2U_m \sqrt{F}}{\left[\left\{ \rho_0 \left[1 + \frac{2}{3} \alpha (T_1 - T_0) \right] \right\}^2 \pi (0.1 H_0) + 4.45 \times 10^{-7} \times 4U_m^2 \right]^{\frac{1}{2}}} \quad \text{EQ. 10}$$

where ρ_0 is initial material resistivity of the contact, T_1 is welding temperature, T_0 is the initial temperature, H_0 is initial hardness of the contact material, and F is contact force. For a remote circuit breaker, the estimated minimum welding current is approximately 2700 A. As such, the contacts are likely to weld in high inrush currents test where the current is approximately 6-7 kA.

Slot motors are generally used to prevent chattering and excess heat generation in the secondary contacts in a circuit breaker as shown in FIG. **2**. FIG. **2** shows a circuit breaker **10** including, among others, a conventional slot motor **1000** coupled to a moving arm **210**, secondary contacts **200, 220**, a solenoid **300**, an actuator **400**, and a spring **500**. The actuator **400** may be of plastic, and the moving arm **210** may be of copper. The slot motor **1000** may be of carbon steel.

FIGS. **3A-C** illustrate an example conventional slot motor **1000**. FIG. **3A** is a cross-sectional view of the slot motor

1000, FIG. 3B is a side view of the slot motor **1000** attached to the moving arm **210**, and FIG. 3C is a perspective view of the slot motor **1000** attached to the moving arm **210**. The conventional slot motor **1000** includes a top slot motor component with, e.g., 0.062 inch nominal thickness and a bottom slot motor component **1020** with, e.g., 0.062 inch nominal thickness. The top slot motor component **1010** has a U-shape with a base **1012** and a pair of legs **1014,1015** extending downward towards the bottom slot motor component **1020**. There are vertical gaps **1018A,B** between each leg **1014,1015** of the top slot motor component **1010** and the bottom slot motor component **1020**, and horizontal gaps **1017A,B** between the moving arm **210** and the inner surface of the pair of legs **1014,1015** of the top slot motor component **1010**. If an inrush current passes through the moving arm **210** of the secondary contacts in contact, magnetic fields will be generated, creating magnetic force (slot motor force) at the contacts **200,220**. The slot motor force $F_{slot\ motor}$ increases inversely to the size of the gaps **1017A,B** and **1018A,B** (assuming no deformations in the contacts **200,220**).

FIG. 4 shows opening strength of the circuit breaker **10**. The contacts **200,220** in contact are impacted by opening force F_{open} **240** and weld force F_{weld} **242**, and solenoid force $F_{solenoid}$ **244** pulling the actuator (e.g., an actuator **400** of FIG. 2) to lift the moving arm **210** to open the contacts **200,220**. If opening force F_{open} is greater than the tack weld force F_{weld} , the tack weld can be opened. Contact opening force is provided as:

$$F_{open} = \alpha_{lever} \beta_{peel} F_{solenoid} \quad \text{EQ. 11}$$

where α_{lever} is the lever arm ratio (e.g., approximately 0.5), β_{peel} is floating pivot peel effect (e.g., approximately 3), and $F_{solenoid}$ is the solenoid force. The maximum weld strengths can be estimated and FIGS. 5A-C show the estimated maximum weld strengths using the conventional slot motor **1000**. The estimated maximum weld strengths were obtained using measured solenoid opening forces as enumerated in the solenoid configurations as shown in Table 1 below.

TABLE 1

	Solenoid Configurations			
	Force at handle		Force at weld	
	lbf	N	lbf	N
Solenoid I (25 V)	1.3	5.8	8.7	
Higher Drive (50 V)	3.3	14.7	22.0	
Solenoid II (50 V)		23.0	34.5	

FIG. 5A shows the estimated maximum weld strengths at 25V drive with opening forces 5.8 N (1.3 lbf) at handle and 8.7 N at weld. FIG. 5B shows the estimated maximum weld strengths at 50V drive with opening forces 14.7 N (3.3 lbf) at handle and 22.0 N at weld. FIG. 5C shows the estimated maximum weld strengths at 50V drive with opening forces 23.0 N at handle and 34.5 N at weld, using a different solenoid (solenoid II). However, it has been shown that at 25V drive using the conventional slot motor **1000**, approximately 15% of welds could be opened. With 50V drive using the conventional slot motor **1000**, about 80% of welds could be opened. With 50V drive using a different solenoid, the conventional slot motor **1000** struggled to always open the welds. A simulation model predicted about 60% weld opening at 25V drive using the conventional slot motor **1000**, 75% weld opening at 50V drive, and about 90% weld

opening at 50V drive with a different solenoid. While the model may not be highly accurate, it gives qualitative agreement to the estimated weld strength described above.

FIGS. 6A-B are internal views of a circuit breaker **20** with the inventive slot motor **2000** according to an example embodiment of the disclosed concept. The circuit breaker **20** is structured to be connected to a power source via a line conductor and a load via a load conductor, and includes primary contacts **100**, a trip mechanism **120**, an operating mechanism **140** structured to cause the primary contacts **100** to open to interrupt current flowing to the load upon detecting a high current event, the inventive slot motor **2000** and secondary contacts **200,220**. The circuit breaker **20** also includes a solenoid **300**, an actuator **400**, a first spring **500**, and a second spring **600**. The circuit breaker **20** may be remotely-operable by an end-user or the utilities via wireless (e.g., WiFi, Bluetooth®, LTE, etc.) or wired connections. The secondary contacts (a movable contact **200** attached to a moving arm **210** and a stationary contact **220** attached to a stationary arm **230**) act as a remote switch for the circuit breaker **20** by the end-user. The slot motor **2000** includes a top slot motor component **2010** and a bottom slot motor component **2020**, and is structured to prevent chattering and excess heat generation in the secondary contacts **200,220** in the circuit breaker **20**. The top slot motor component **2010** is structured to be attached (e.g., via riveting, gluing, welding, etc.) to the moving arm **210**. The bottom slot motor component **2020** has a U-shape with a base and a pair of legs. The U-shaped bottom slot motor component **2020** is structured to remain stationary. It is held in place to the plastic housings of the circuit breaker **20** with the slots **2026** on the external side surfaces of the pair of legs. It may include holes in the sides (the pair of legs) with corresponding pegs in the enclosure body. The top and bottom slot motor components **2010,2020** are separated from each other by a vertical gap between the bottom surface of the top slot motor component **2010** and the top portions of the pair of legs of the bottom slot motor component **2020**. When current passes through the moving arm **210**, the top and bottom slot motor components **2010,2020** generate a magnetic field, which creates a closing force to maintain the secondary contacts **200,220** closed. The second spring **600** is structured to push on a pin within the solenoid **300**, thereby pushing the actuator **400** down. The actuator **400**, in turn, pushes on the moving arm **210**. The additional torsion spring **600** provides additional closing force for the secondary contacts **200,220**. The slot motor **2000** is discussed further in detail with reference to FIGS. 7A-E. FIG. 6B shows the current path **250** in the circuit breaker **20** when the secondary contacts **200,220** are closed. It starts from a utility power source (not shown) via a source terminal and ends at a load (not shown).

FIGS. 7A-E illustrate a slot motor **2000** according to an example embodiment of the disclosed concept. FIG. 7A illustrates a cross-sectional view of the slot motor **2000** including a top slot motor component **2010** attached (e.g., riveted, welding, gluing, etc.) to the moving arm **210** and a bottom slot motor component **2020**. Slot motor **2000** is different from the conventional slot motor **1000** in several ways. First, the top slot motor component **2010** is no longer U-shaped as in the conventional slot motor **1000**, and the bottom slot motor component **2020** now has the U-shape, including a base **2022** and a pair of legs **2024,2025** extending upwards from the base **2022** toward the top slot motor component **2010**. This configuration places the vertical gaps **2018A,2018B** much closer to the moving arm **210** as compared to those in the conventional slot motor **1000**, thereby

increasing the magnetic fields generated by the slot motor in the event of an inrush current.

Further, this configuration allows the thickness of the bottom slot motor component **2020** to be increased at the base **2022** and the pair of legs **2024,2025**. This is important in that the magnetic field is proportional to the current. However, as the current and magnetic field increase, the blow-off force $F_{blow\ off}$ also increases. Therefore, in order to increase forces generated by the slot motor **2000** without increasing $F_{blow\ off}$, increasing the thickness of the top and bottom slot motor components is critical. Given the limited space within the circuit breaker, inverting the slot motor (upside down) achieves the increase in thickness of the slot motor **2000** as desired. As shown in FIG. 7A, the top slot motor component **2010** has, e.g., without limitation 0.062 inch in nominal thickness **2030** for the area **2031** in which the actuator **400** is placed and 0.093 inches in nominal thickness **2032** for portions surrounding the area **2031** in which the actuator **400** is placed. This is a significant increase in thickness of the top slot motor component **2010** as compared to the 0.062 inch nominal thickness all around for the bottom slot motor component **1020** of the conventional slot motor **1000** as discussed with reference to FIG. 3A. Likewise, for the bottom slot motor component **2020** the height **2034** of the pair of legs **2024,2025** is, e.g., without limitation, 0.203 inch and the base **2022** has varying thicknesses (e.g., without limitation, 0.055 inches in nominal thickness **2036** for the area **2035**, 0.115 inches in nominal thickness **2038** around the area **2035**, etc.) as shown in FIG. 7B. This is also a significant increase in thickness over the top slot motor component **1010** having the nominal thickness of 0.062 inches for the base **1012** and the pair of legs **1014,1015** of the conventional slot motor **1000**. In some examples, as a result of the increased thickness of the slot motor **2000**, chamfers and rounds to both the top and bottom slot motor components **2010,2020** are added in order to make the slot motor **2000** fit in the circuit breaker as shown in FIGS. 8A-B.

Second, the vertical gaps **2018A,B** are made as small as possible and placed in the same plane as the moving arm **210** is placed in order to also increase the magnetic field, and thus, increase the closing force generated by the slot motor **2000**. The slot motor **2000** is made of magnetic material, e.g., magnetic steel, with high saturation flux density and low coercivity. FIG. 7C is a side view of the slot motor **2000** with the moving arm **210** attached to the top slot motor component **2010**. The top slot motor component **2010** is structured to be attached to the moving arm (e.g., a remote contact moving arm for a smart circuit breaker) **210**. The bottom slot motor component **2020** is stationary. FIG. 7D is an exploded view of the slot motor **2000**. The top slot motor component **2010** includes connecting elements **2012** (e.g., screw holes) structured to connect the top slot motor component **2010** with the moving arm **210**. FIG. 7E illustrates a moving arm **210** fit to the bottom slot motor component **2020** and the bottom slot motor component **2020** having varying widths **2037** (e.g., without limitation, 0.094 nominal inches) and **2039** (e.g., without limitation, 0.123 nominal inches).

FIGS. 8A-B illustrate secondary contacts in different positions according to an example embodiment of the disclosed concept. FIG. 8A is a side view of the slot motor **2000** with the secondary contacts **200,220** in a closed position. The moving arm **210** pivots at location **260** in the plastic housing of the circuit breaker **20**. FIG. 8B is a perspective

view of the slot motor **2000** with the secondary contacts **200,220** in an open position.

FIGS. 9A-D illustrate the magnetic fields **2040** and closing force (slot motor force) **2042** generated by the slot motor **2000,2000'** according to an example embodiment of the disclosed concept. When current passes the moving arm **210**, the top and bottom slot motor components **2010,2020** generate magnetic fields. These magnetic fields create a force **2042** to keep the contacts **200,220** closed (in contact) as shown in FIG. 9A. FIG. 9B shows magnetic fields generated by the slot motor **2000'** having 0.2 mm horizontal gaps **2017A,B** between the pair of legs **2024',2025'** of the bottom slot motor component **2020'** and the top slot motor component **2010'**. The slot motor **2000'** is different from the slot motor **2000** in that it has the same thickness for both the top and bottom slot motor components **2010'** and **2020'** all around. In FIG. 9C, the slot motor **2000'** generates more $F_{slot\ motor}$ **2044** than the $F_{slot\ motor}$ **2043** generated by the conventional slot motor **1000**. FIG. 9D shows that there is more than 60% increase in the $F_{slot\ motor}$ at the maximum current. It has been shown that the slot motor force $F_{slot\ motor}$ depends rather weakly on the gaps between the top and bottom slot motor components **2010,2020**.

FIGS. 10A-B illustrate reduced slot motor force due to horizontal misalignment in a slot motor **2000'** according to an example embodiment of the disclosed concept. When the horizontal gaps **2017'A** (0.1 mm) and **2017'B** (0.3 mm) between the moving arm **210** and the pair of legs **2024',2025'** of the bottom slot motor component **2020'** is misaligned as shown in FIG. 10A, the horizontal force **2046** is reduced to 1.3 N (0.31 bf) as shown in FIG. 10B from, e.g., the horizontal force of the slot motor **2000'** with the aligned horizontal gaps **2017A,2017B** (i.e., both being 0.2 mm).

FIGS. 11A-D illustrate changes in slot motor force according to an example embodiment of the disclosed concept. FIG. 11A-D show that the inventive slot motor **2000'**, which is inverted in shape as compared to the conventional slot motor **1000**, has the highest slot motor force **2044**, the conventional slot motor **1000'** with a vertical offset for the moving contact **210** has the second highest slot motor force **2045**, the conventional slot motor **1000"** with split moving arms (two moving arms) **210'** has the third highest slot motor force **2047**, and the conventional slot motor **1000** has the lowest slot motor force **2043**. The vertical offset means that the moving arm **210** is vertically moved down towards the bottom of the top U-shaped slot motor component **1010'** of the conventional slot motor **1000'**.

FIGS. 12A-B illustrate magnetic field and force generated by a slot motor **2000"** according to an example embodiment of the disclosed concept. As shown in FIG. 12A, the base **2022"** and a pair of legs **2024",2025"** have increased thicknesses than those of the slot motor **2000,2000'**. FIG. 12B shows that the force **2048** generated by the slot motor **2000"** is double the amount of force **2044** generated by an example inverted slot motor **2000'** having the same thickness (e.g., 0.062 inches) as the conventional slot motor **1000**, and almost three times more than the amount of force **2049** generated by a conventional slot motor having 0.3 mm thickness.

FIGS. 13A-C shows balance of forces associated with the secondary contacts **200,220** in a circuit breaker **20** using a slot motor (inverted slot motor) **2000** according to an example embodiment of the disclosed concept. The circuit breaker **20** includes a slot motor **2000** attached to the moving

arm **210**, a solenoid **300**, and a spring. With the secondary contacts **200, 220** in contact, FIG. **13A** shows a balance of forces based on the solenoid force $F_{solenoid}$ **244'** being applied to an actuator for the moving arm **210**, blow off force $F_{blow\ off}$ **241** and blow on force $F_{blow\ on}$ **243** at the contacts **200, 220**, and spring force F_{spring} **245** pressing down on the movable contact **210**. The blow off force $F_{blow\ off}$ **241** attempts to separate two contacts **200, 220** when the current is flowing between them when they are in contact. The blow on force $F_{blow\ on}$ **243** attempts to keep the two contacts **200, 220** remain in contact. FIG. **13B** shows the net forces **250** at the contacts **200, 220** using the inverted slot motor **2000** is significantly higher than the net forces **252** at the contacts using the conventional slot motor **1000**. The higher net forces at the contacts **200, 220** with the inverted slot motor **2000** result in less contact resistance and less energy E_{melt} to melt the contact material than those when using the conventional slot motor **1000**. FIG. **13C** shows that relative contact resistance decreases as the inrush current increases with the use of the inverted slot motor **2000**.

FIG. **14** shows contact forces for no tack weld according to an example embodiment of the disclosed concept. FIG. **14** illustrates minimum contact force **254** for no weld, contact force **251** for a circuit breaker using the inverted slot motor **2000**, and contact force **253** for a circuit breaker using the conventional slot motor **1000**. As shown in FIG. **14**, the contact force **251** is greater than the minimum for inrush currents less than 4000 A and the contact force **253** is greater than the minimum for inrush currents less than 2500 A. Thus, for a circuit breaker using the inverted slot motor **2000**, there may be no welds for currents less than 4000 A. For a circuit breaker using the conventional slot motor **1000**, the contact force **253** is greater than the minimum for inrush currents less than 2500 A. It has been observed that for a circuit breaker using the slot motor **2000** and an arc bypass of 3500 A, there may be no welds for currents less than 7500 A. In order to avoid welding in an arc bypass, the current flowing in the contacts $I_{contacts}$ needs to be less than the minimum welding current as shown below:

$$I_{contacts} = I - I_{bypass} < I_{min\ weld} \quad \text{EQ. 14}$$

where I_{bypass} is the arc bypass current. As such, if there is an arc bypass when using the inverted slot motor **2000**, then to achieve no weld it needs to siphon off about 3500 A for the maximum inrush current of 7500 A.

FIG. **15** illustrates relative weld strength of the slot motor **2000** as opposed to the conventional slot motor **1000** according to an example embodiment of the disclosed concept. The maximum weld strengths with the inventive slot motor **2000** and the conventional slot motor **1000** are estimated. With the use of the inventive slot motor **2000**, the maximum weld strength may be reduced to 25%-50% of the conventional slot motor **1000**.

While specific embodiments of the invention have been described in detail, it will be appreciated by those skilled in the art that various modifications and alternatives to those details could be developed in light of the overall teachings of the disclosure. Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of disclosed concept which is to be given the full breadth of the claims appended and any and all equivalents thereof

What is claimed is:

1. A slot motor for use with secondary contacts in a circuit breaker, comprising:

a movable top slot motor component structured to be directly attached to a moving arm of the secondary contacts; and

a stationary U-shaped bottom slot motor component including a base and a pair of legs extending upward from the base, the U-shaped bottom slot motor component structured to be separated from the top slot motor component by vertical gaps between the top slot motor component and ends of the pair of legs,

wherein the slot motor is structured to generate a magnetic field producing a force to maintain the secondary contacts in a closed position during a high current event.

2. The slot motor of claim 1, wherein the moving arm is disposed within the U-shaped bottom slot motor component.

3. The slot motor claim 1, wherein the vertical gaps are in the same plane as a plane in which the moving arm is arranged.

4. The slot motor of claim 1, wherein the vertical gaps are minimized to increase the magnetic field and the force.

5. The slot motor of claim 1, wherein the ends of the pair of legs of the U-shaped bottom slot motor component are separated from the moving arm by horizontal gaps.

6. The slot motor of claim 5, wherein the horizontal gaps have a same size.

7. The slot motor of claim 1, wherein thicknesses of the top slot motor component and the U-shaped bottom slot motor component are maximized to increase the magnetic field and the force.

8. The slot motor of claim 7, wherein the top slot motor component includes a center portion in which an actuator for the moving arm is arranged and a remaining portion around the center portion, and the center portion of the top slot motor component has a first thickness and the remaining portion has a second thickness larger than the first thickness.

9. The slot motor of claim 7, wherein the base of the U-shaped bottom slot motor component comprises a first thickness at a center portion surrounding an end of the actuator and a remaining portion around the center portion, and the center portion has a first thickness and the remaining portion has a second thickness larger than the first thickness.

10. The slot motor of claim 1, wherein the bottom slot motor component is held in place to a housing of the circuit breaker via slots on external side surfaces of the pair of the legs.

11. The slot motor of claim 1, further comprising one or more chamfers to at least one of the top slot motor component or the U-shaped bottom slot motor component, the one or more chamfers structured to make the slot motor to fit within the circuit breaker.

12. The slot motor of claim 1, wherein the top slot motor component and the U-shaped bottom slot motor component are made of soft magnetic materials.

13. The slot motor of claim 1, wherein the secondary contacts act as a remote switch for the circuit breaker.

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